

PROBLEM FRAMING CANVAS

STAKEHOLDER & DECISION MAKER



The bank's marketing/telemarketing operations team, running term-deposit outbound campaigns.

UNIT OF ANALYSIS



One client, represented by their most recent campaign contact record.

TASK TYPE



Supervised binary classification.

DECISION NEED



Which existing clients should be prioritized for the next round of outbound calls, given limited agent time and a real cost per call.

PRIMARY LENS



Campaign response prediction — predict whether a client will subscribe (y) to the term deposit.

SECONDARY LENS



Customer segment prioritisation — convert each client's predicted probability into a priority tier (High / Medium / Low) for call allocation.

TARGET / OUTPUT



Predicted probability that a client subscribes ($y = \text{yes}$); used to rank and tier clients for calling priority.

KEY INPUTS



Client demographics, financial status, and campaign contact history. duration is deliberately excluded — confirmed leakage (only known after a call takes place).

SUCCESS MEASURE



Business-relevant ranking quality (PR-AUC, Lift@Top-K) and expected profit under a calling-cost assumption — not raw accuracy, given ~88/12 class imbalance.

CONSTRAINTS & RISKS



Severe class imbalance; structural missingness in outcome; leakage risk via duration; data era (~2008–2010) may not reflect current behaviour; fairness review of age/marital not yet performed.

RATIONALE



Turns a historical dataset into a forward-looking operational tool — gives the bank a defensible, evidence-based way to decide who to call next, reducing wasted agent time and enabling evidence-based resource allocation.